

Praktische Übung zur Vorlesung Seismologie I Abgabe Dienstag 28.4.26 bis 12.00

Hochladen auf learnweb (Antworten zu Aufgaben 3,4,5)

Aufgaben, die nach 12 Uhr aber vor Mitternacht am Dienstag (28.4.2026) abgegeben wurden, verlieren 20% der Punkte, Übungszettel, die am Mittwoch 29.4.2026 abgegeben wurden verlieren 40% der Punkte. Übungszettel, die nach Mittwoch 24Uhr abgegeben werden, werden nicht mehr korrigiert und bekommen keine Punkte.

At home:

Use also the available literature for answering the questions

3) If you have a square 2D volume (x_1, x_2 direction) element, show what the strain element

$$\frac{\partial u_2}{\partial x_2} < 0, u_1 = 0 \quad \text{and} \quad \frac{\partial u_1}{\partial x_2} > 0, \frac{\partial u_2}{\partial x_1} < 0 \quad \text{would do to this 2D volume?}$$

4) a) For the strain tensor

$$\varepsilon = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 2 \end{pmatrix}$$

find the corresponding stress tensor σ , assuming an isotropic material with the two Lamé parameters, λ, μ .

b) Find the stored strain energy: $W = \sigma_{ij} \varepsilon_{ij} / 2$

5) Does Hooke's law hold for all magnitudes of applied force? Explain your answer.

Practical Exercise (in CIP room during Practical)

6) This practical exercise is supposed to remind you about reading data into seismic handler, and detecting seismic arrivals. There is a seismic handler command file on learnweb!
In your practical work, always record your observations and tests (you might need this later, in courses or even in your future work).

copy data from /scratch:

`cp -r /scratch/cthomas/SEISMO1/PRAC2 .` (the dot is important at the end)

there are 2 files: 11MAR03.QHD and 11MAR03.QBN. This event is from 11th of March 2003 7:51:35 in the Papua New Guinea Sea, recorded in Germany, at stations of the GRSN

Find the event on the NEIC web site <https://earthquake.usgs.gov/earthquakes/search/>

Loading data into seismic handler:

open Seismic Handler (*shc7.0*)

and open a window *xopen;;*

check the seismic handler documentation...(also on learnweb)

Read data into seismic handler `read 11mar03 all`

Check whether all information is correct (lat/lon/depth/origin)

`list_all lat`

`list_all lon`

etc (for all parameters such as origin, lat, lon, depth, etc) (look the correct values up on the NEIC web site)

to set distance and backazimuth, type

`epi_dist ^(lat1) ^(lon1)`

P- and S-waves

Rotate event into ZRT (`rot_all_old`) after making sure they are sorted ZNE before (if not, type `sort_zne` but make sure there are 3 components of each station).

P-waves are usually on the Z component, SH on the T and SV on the R component.

You can look at different components individually

`hide _comp~(z)` for example means: hide all components that are not Z.

To get all back: `display h:all`

Try to identify the arrivals on the components. Try first without help, then use `mark_taup;;` which gives you the main arrivals.

If you want to remember which waves you are looking at, open *taup* in a new window (just type *taup*). There you can select waves and distance and depth of the earthquake and look at either travel time/slowness etc or paths. Try a few waves (e.g. PKiKP, SKS, ScS, PKIKP, PP, P,)

Make screenshots and write a short report of what you did (max 1 page) including the pictures.